



Convex Optimization

Project 5



Spring 1403

Due date: 4th of Khordad

1. *Optimal certificate of deposit investment.* You have an initial endowment of cash $C_1 > 0$ that you will use to pay a series of known liabilities $l_t \geq 0$ in periods $t = 1, \dots, T$. We will assume that $C_1 \geq \sum_{t=1}^T l_t$, *i.e.*, the initial cash can cover all the liabilities.

In any period you can invest a nonnegative amount in CDs (certificates of deposit, also known as zero-coupon bonds). A CD is characterized by its maturity $M \geq 1$ (given in periods) and its (per period) interest rate $R > 0$. If you invest an amount $z \geq 0$ in such a CD in period t , you pay $z \geq 0$ in period t (to fund the CD), and you will receive the amount $z(1+R)^M$ in period $t+M$. We say that the CD matures in period $t+M$. (Here we assume that $t+M \leq T$.)

There are K different CDs available, with maturities and rates given by M_k and R_k , $k = 1, \dots, K$. (The maturities are distinct, *i.e.*, there is at most one CD with a given maturity.) Let $z_{tk} \geq 0$ denote the amount invested in CD k in period t , with $z_{tk} = 0$ when $t+M_k > T$. (That is, we do not invest in CDs which mature after time period T .) The total amount you must pay in period t to fund these CDs is

$$F_t = \sum_{k=1}^K z_{tk}$$

The amount you receive in period t from previous CDs maturing is

$$G_t = \sum_{\tau+M_k=t} z_{\tau k} (1+R_k)^{M_k}$$

The sum here is over all $\tau \in \{1, \dots, t-1\}$ and k for which $\tau + M_k = t$, *i.e.*, over all CDs that mature in period t .

The cash dynamics are given by

$$C_{t+1} = C_t - l_t - F_t + G_t, \quad t = 1, \dots, T$$

We require that $C_t \geq 0$ for all t . Our objective is to maximize C_{T+1} , the amount of cash on hand after paying the last liability.

The variables in the problem are z_{tk} , $t = 1, \dots, T-1$. The data is C_1 , the liability stream $l_1, \dots, l_T$, and the CD maturities and rates M_k, R_k , $k = 1, \dots, K$.

- (a) Explain how to solve this problem using convex optimization. If you use any change of variables or a relaxation, explain.
- (b) Find optimal CD investments for the problem instance with data in `cd_invest_data.py`. The data consists of 120 monthly liabilities over 10 years, and $K = 6$ CDs with maturities of 1, 3, 6, 12, 24, and 36 months, and interest rates of 1%, 2%, 3%, 4%, 5%, and 6% annually. (These are converted to per-period, *i.e.*, monthly, interest rates $R_1, \dots, R_6$ in the data file.) Report the optimal final cash value C_{T+1}^* for your investments. Plot z_{tk}^* versus t for each M_k , on different plots. Compare the optimal policy to a baseline policy that invests all cash in each period in the CD with a maturity of $M = 1$.

Remarks. (Not needed to solve the problem.) We are making a number of simplifying assumptions that do not hold in practice. In practice (but not in this simplified problem):

- The interest rates for a given maturity CD change over time.
- The liabilities are not known, but estimated.
- You can redeem a CD before maturity, with a penalty such as a reduced interest rate.

Good Luck!